

KEERTHI SA

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Machine Learning Engineer | Agentic Systems | Semantic Retrieval | LLM Serving

PROFESSIONAL SUMMARY

Machine learning engineer with 4+ years building and shipping applied ML and large language model (LLM) systems at Amazon and Google. Focused on agentic workflows, embedding-based semantic retrieval, and high-throughput model serving, with a track record of turning engineer and customer pain points into production tools. Owns ML systems end to end, from data pipelines and model optimization through evaluation and deployment.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, SQL, C++

ML and Generative AI: PyTorch, Transformers, PEFT/LoRA, fine-tuning, INT4 quantization, RAG, agentic and multi-agent workflows

Retrieval and Search: embedding-based retrieval, FAISS (HNSW), semantic search, MMR reranking

ML Serving and Optimization: speculative decoding, continuous batching, KV-cache/PagedAttention, vLLM, Ray, model evaluation

Data and Pipelines: Spark, Kafka, Airflow, DuckDB, Redis, feature stores, PSI drift detection

Cloud and Platform: AWS (SageMaker, Bedrock, EKS), GCP (Vertex AI), Kubernetes, Docker, Terraform

Foundations: system design, distributed systems, API design, A/B experimentation

PROFESSIONAL EXPERIENCE

Amazon (Seattle, Washington)

Machine Learning Engineer | June 2023 to Present

- Design and operate LLM serving infrastructure processing 1.7B+ inference requests per month, cutting p99 latency 38% through continuous batching and speculative decoding, the same techniques implemented from scratch in my open-source InferenceLab project.
- Build multi-agent workflows using a planner-executor-critic pattern with sandboxed tool execution to automate internal engineering tasks, reducing manual triage time 44%; the architecture mirrors my open-source AgentForge system.
- Develop embedding-based semantic retrieval over a 12M+ document internal corpus, served with FAISS HNSW and MMR reranking, improving top-10 relevance 27%; the retrieval stack is characterized in my open-source TwoTowerRecs project.
- Reduce GPU serving cost \$384K annually by applying INT4 NF4 quantization and LoRA fine-tuning to host 7B-parameter models at 4x lower memory with no measurable quality loss, methods validated in my open-source VisionTalk project.
- Build ML data pipelines and an online/offline feature store with point-in-time correct joins feeding 14 production models, adding PSI-based drift detection that cuts silent model degradation incidents 31%; the design is open-sourced as FeatureFlow.
- Drive design reviews and technical direction for the team's model-serving roadmap and mentor 3 junior engineers on ML systems and evaluation practices.

Google (Hyderabad, India)

Software Engineer | July 2021 to Aug 2022

- Built and shipped ranking features for a recommendation surface serving tens of millions of users, contributing model and pipeline components to a larger team effort.
- Implemented offline evaluation and A/B experiment tooling for candidate-ranking models, improving click-through relevance 9% on the tested segment.

- Developed data preprocessing pipelines that reduced feature-generation runtime 23%, supporting faster model iteration across the team.

EDUCATION

M.S. Information Systems, University of Maryland, Baltimore County | Aug 2022 to May 2024